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(54) **DEVICE, SYSTEM, AND METHOD UTILIZING A RADIOPAQUE ELEMENT FOR ANATOMICAL LESION LENGTH ESTIMATION**

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A61B 90/00 (2016.01)
A61M 25/01 (2006.01)

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(58) **Field of Classification Search**
CPC **A61B 90/06**
See application file for complete search history.

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Primary Examiner — Joseph M Santos Rodriguez

(57) **ABSTRACT**

A catheter is provided with increased flexibility and radiopaque measurement visibility. The radiopaque measurement bands are formed of a continuous coil of radiopaque material defined by areas of tightly packed coils spaced by areas of loosely wound coils. Systems and methods of utilizing the measurement structure are also provided.

12 Claims, 9 Drawing Sheets

Related U.S. Application Data

Jul. 26, 2017, now Pat. No. 10,561,473, which is a continuation of application No. 13/973,773, filed on Aug. 22, 2013, now Pat. No. 9,743,992.

- (60) Provisional application No. 61/692,603, filed on Aug. 23, 2012.

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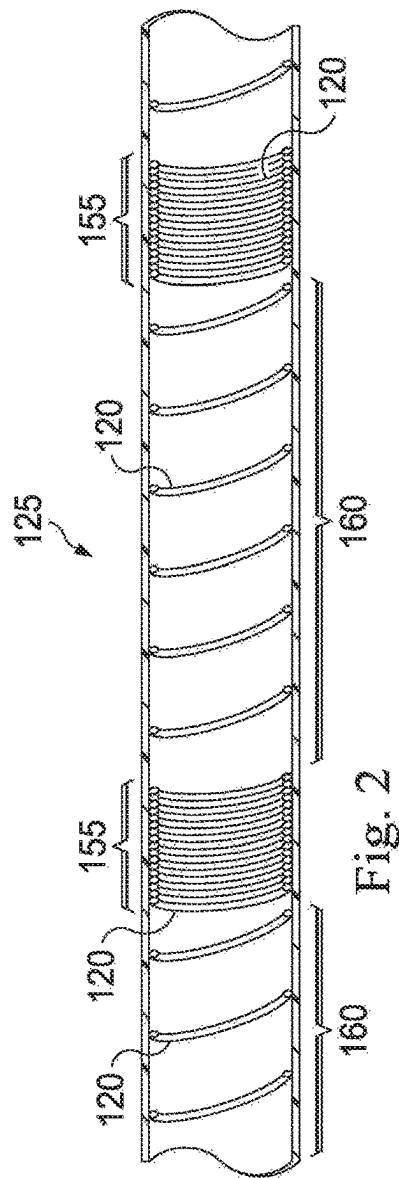
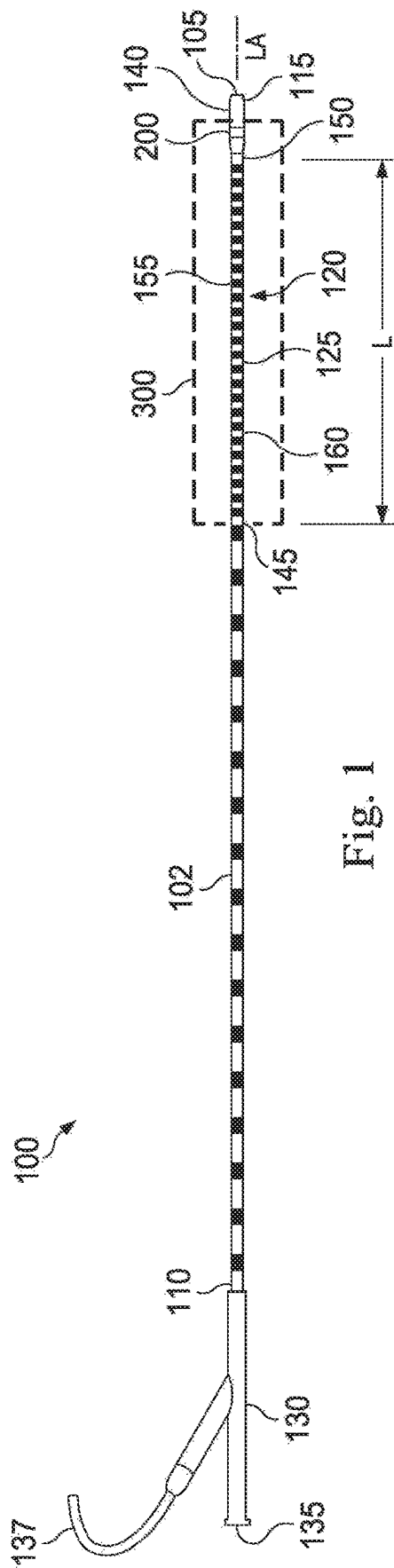
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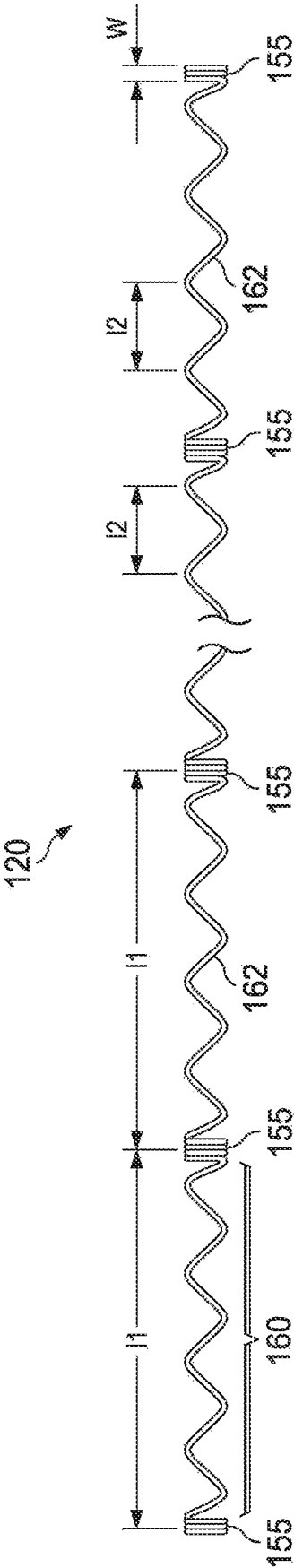


Fig. 3

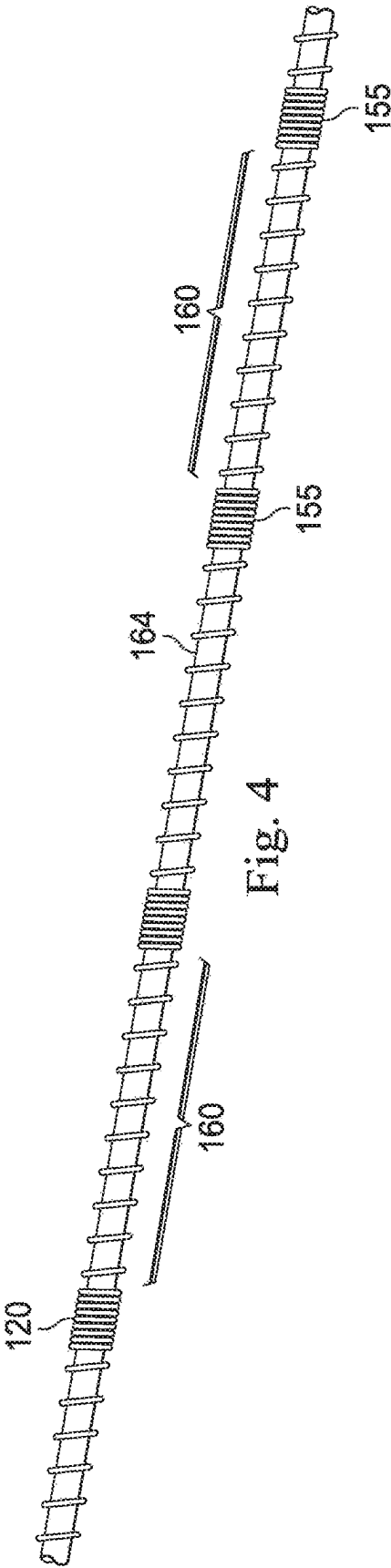


Fig. 4

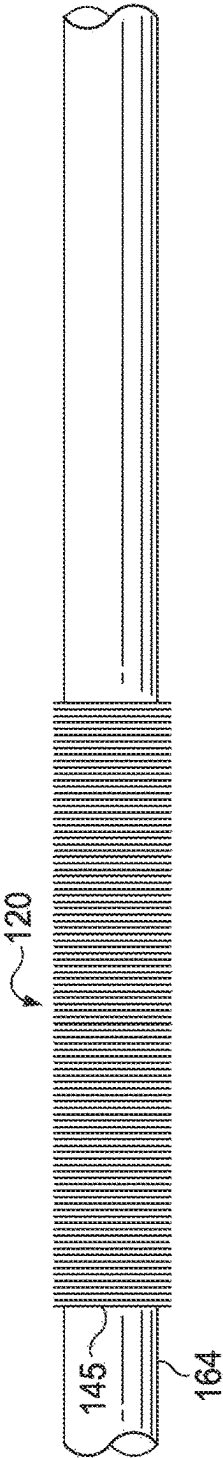


Fig. 5a

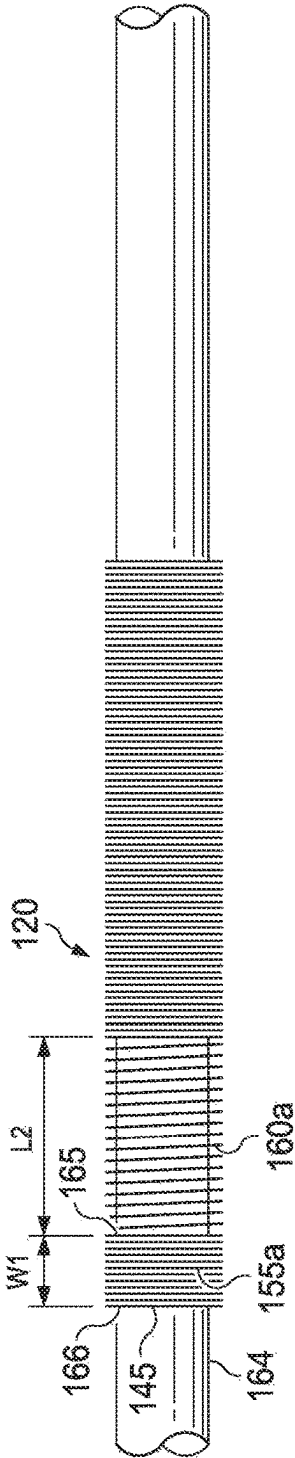


Fig. 5b

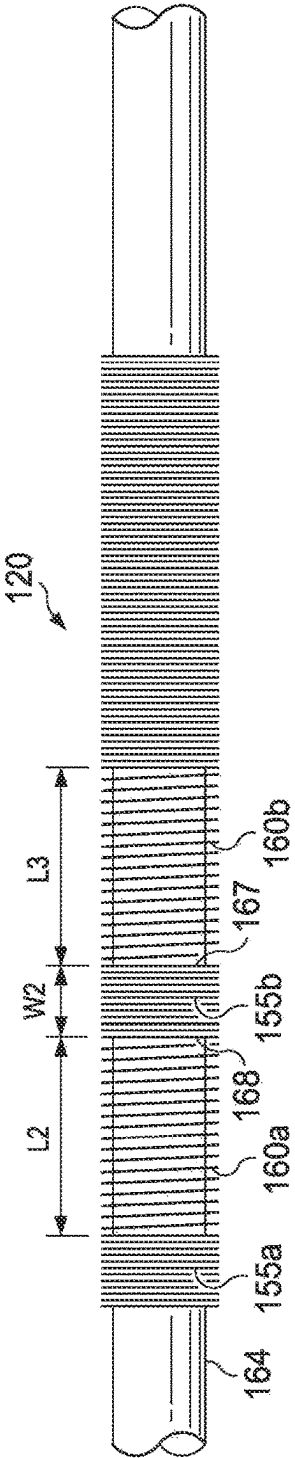


Fig. 5c

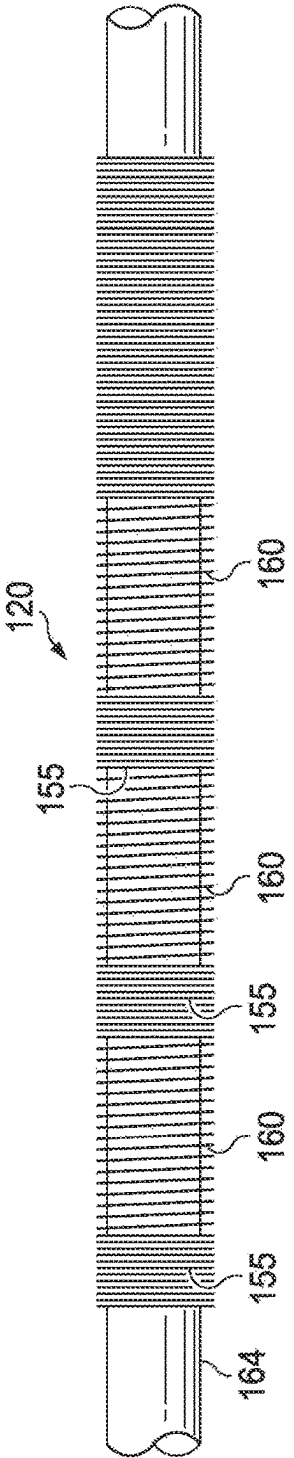
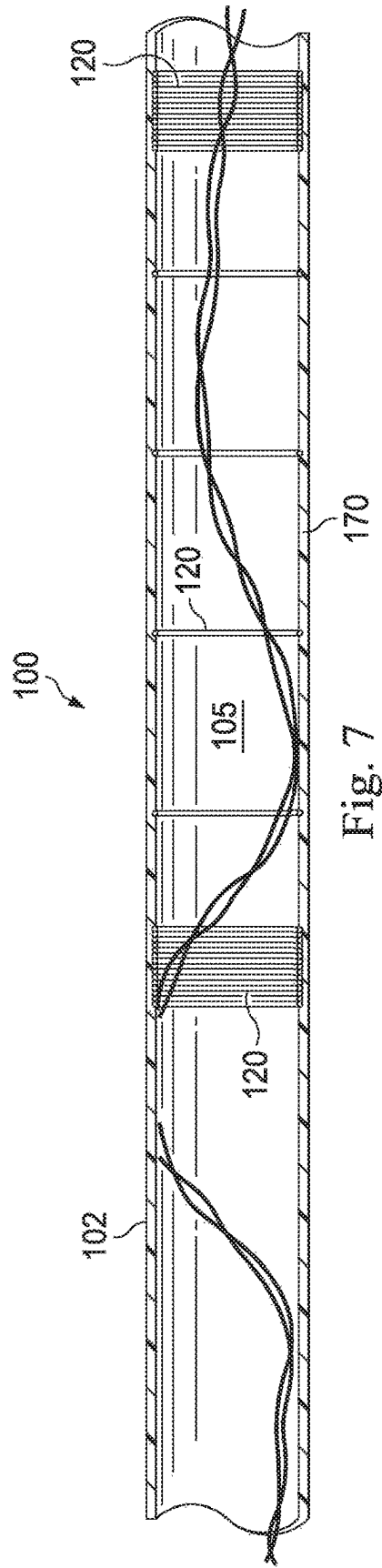
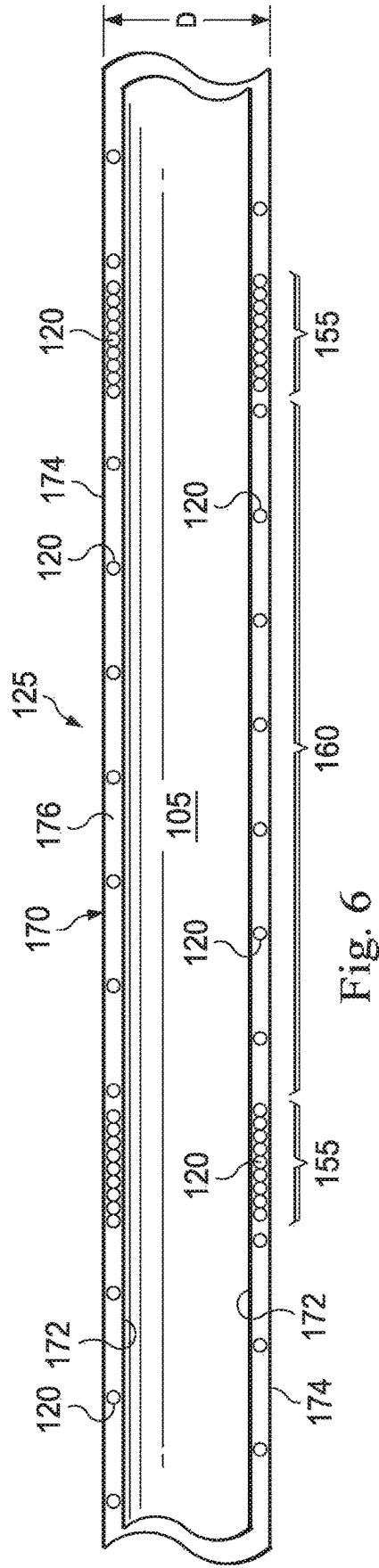


Fig. 5d



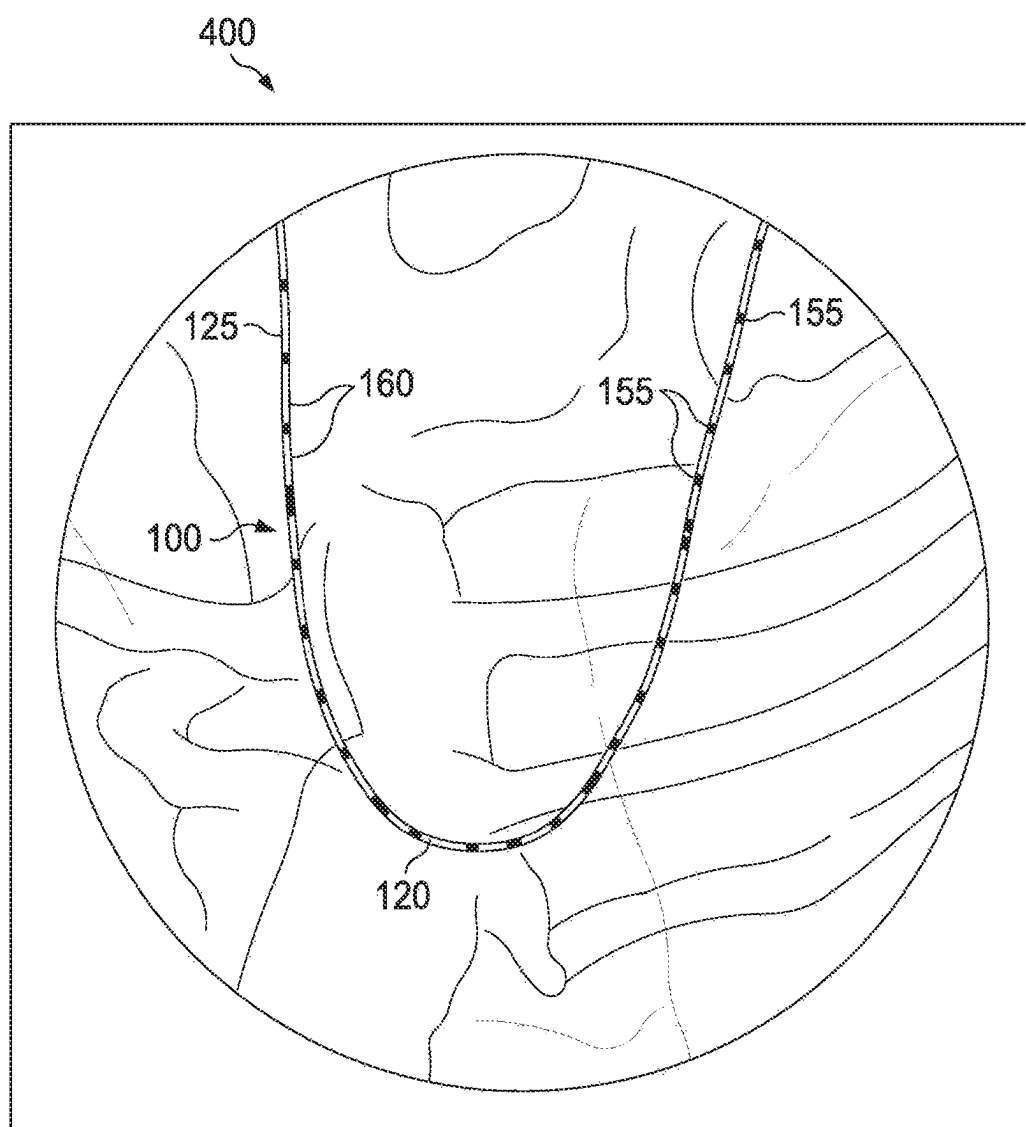


Fig. 8

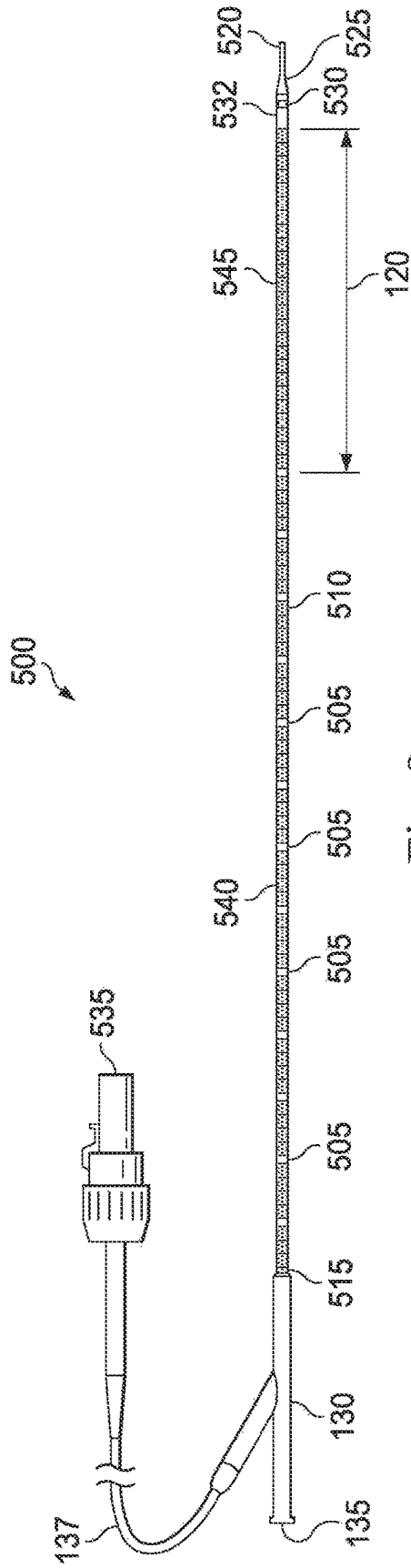


Fig. 9

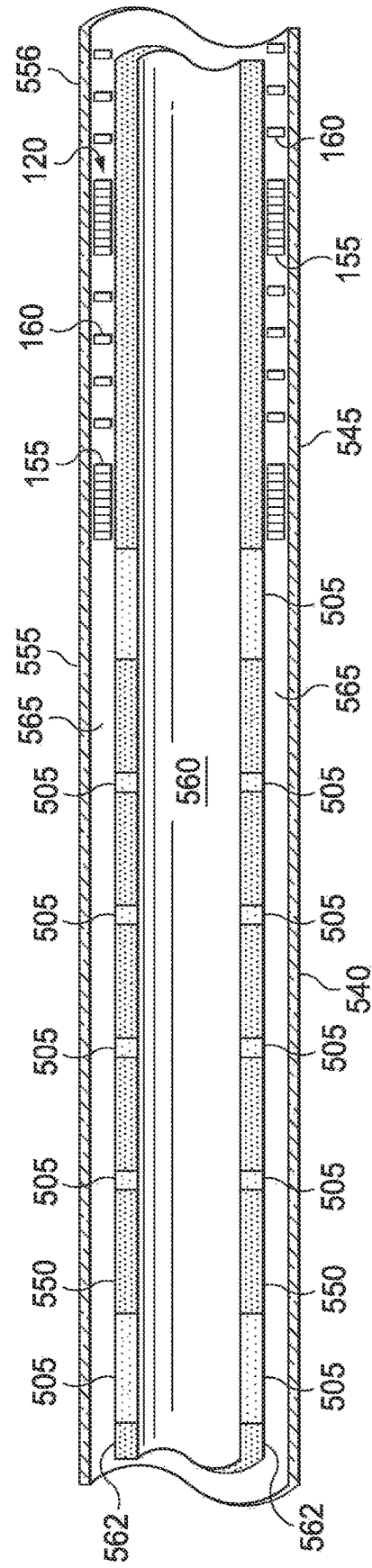


Fig. 10

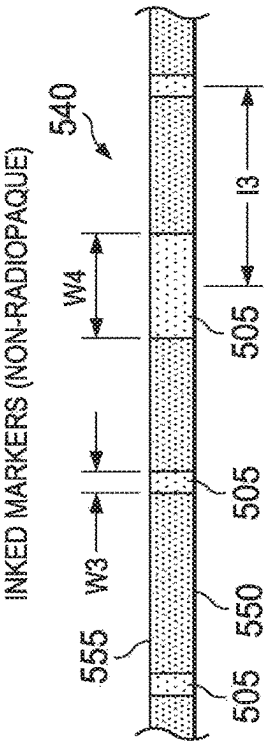


Fig. 11

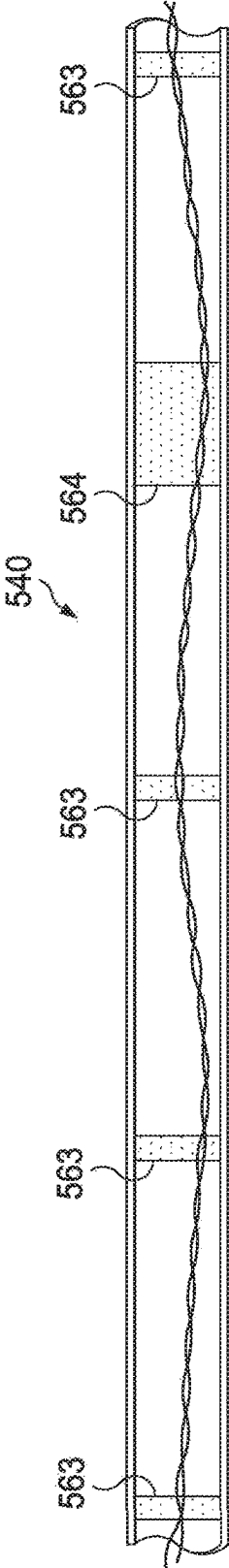


Fig. 12

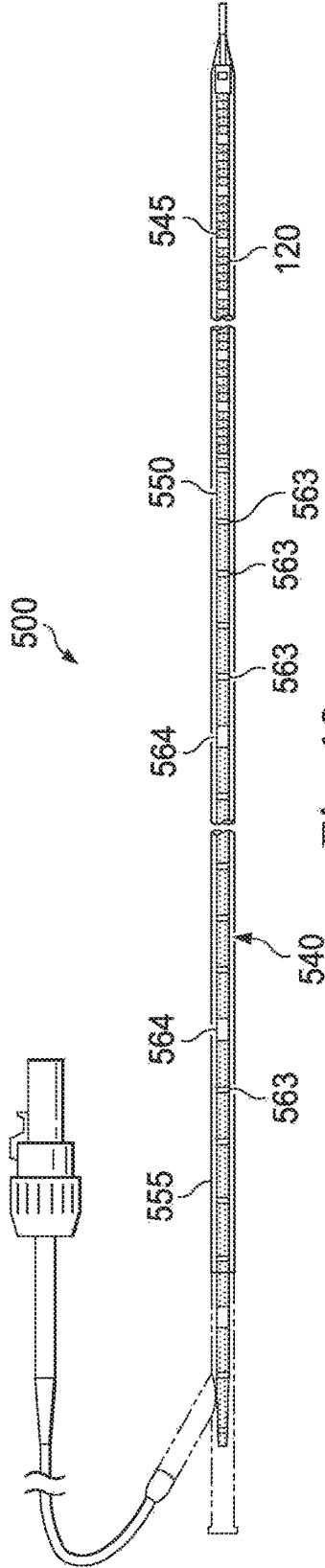


Fig. 13

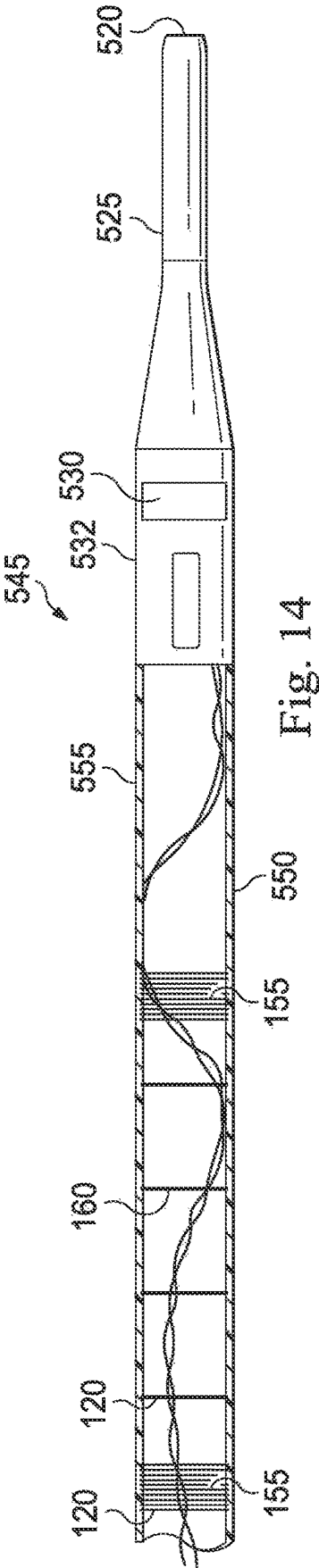


Fig. 14

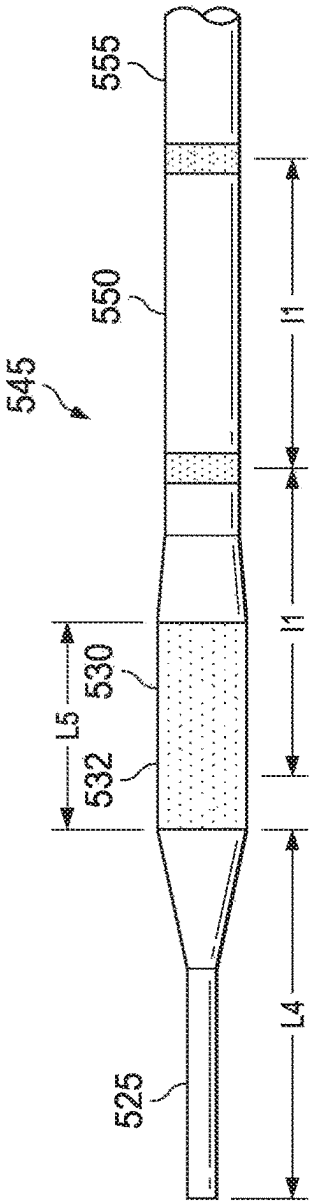


Fig. 15

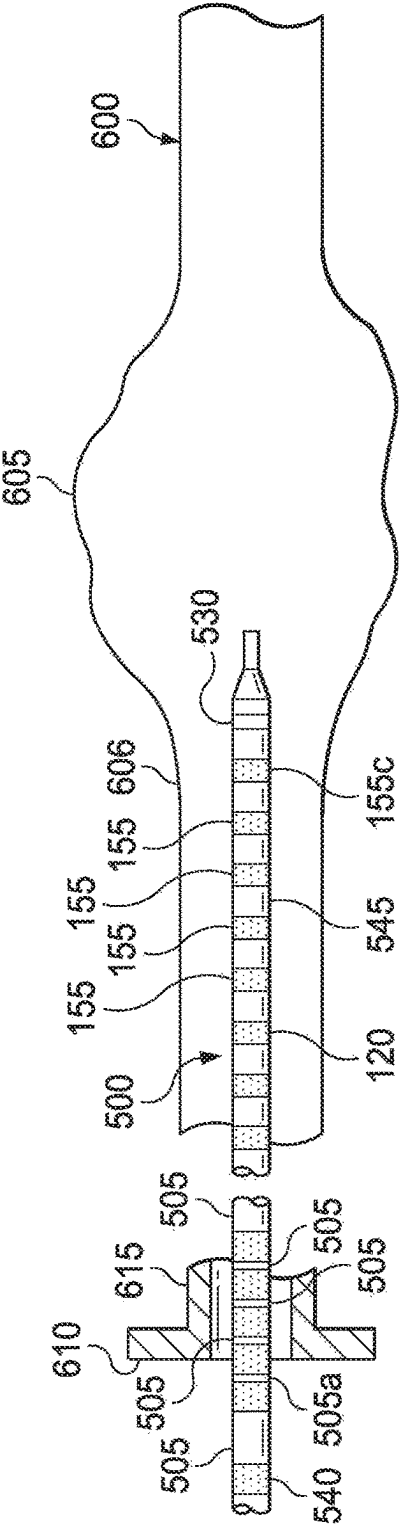


Fig. 16

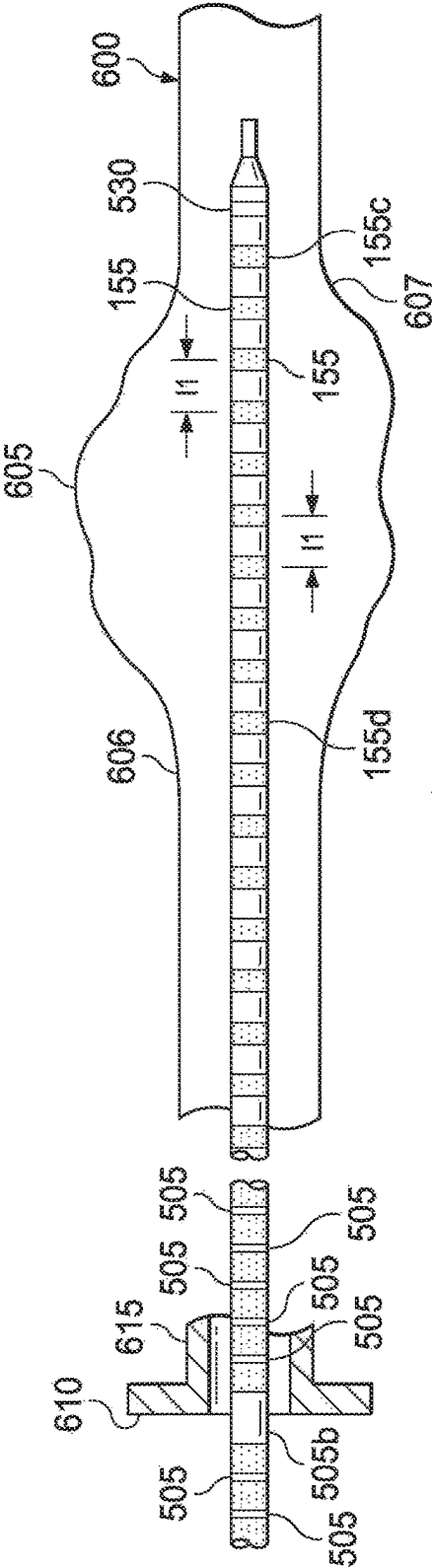


Fig. 17

1

DEVICE, SYSTEM, AND METHOD UTILIZING A RADIOPAQUE ELEMENT FOR ANATOMICAL LESION LENGTH ESTIMATION

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. application Ser. No. 17/739,228, filed May 9, 2022, now U.S. Pat. No. 11,950,965, which is a continuation of U.S. application Ser. No. 16/780,253, filed Feb. 3, 2020, now U.S. Pat. No. 11,850,102, which is a continuation of U.S. application Ser. No. 15/660,446, filed Jul. 26, 2017, now U.S. Pat. No. 10,561,473, which is a continuation of U.S. application Ser. No. 13/973,773, filed Aug. 22, 2013, now U.S. Pat. No. 9,743,992, which claims priority to and the benefit of U.S. Provisional Patent Application No. 61/692,603, filed Aug. 23, 2012, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

Catheters have widespread clinical use in both diagnostic and therapeutic procedures. For example, catheters are used diagnostically to inject contrast media, measure internal body dimensions, retrieve biopsy samples, and visually inspect internal body sites. Catheters are used therapeutically to deliver drugs, deliver implants, drain fluids, retrieve foreign and/or undesirable materials, deliver ultrasound, deliver laser light, provide access for minimally invasive surgical instruments, and dilate narrowed body passages (e.g., vessels).

In several of these procedures, it is advantageous to be able to visualize the progress of the catheter towards the target location within a patient's body. Introducing catheters into the body often requires fluoroscopic visualization to aid the treating healthcare provider in guiding the catheter to the target site. Catheters are commonly formed of a non-radiopaque polymeric material. Therefore, radiopaque markers may be added to the catheter to enable the catheter to be visualized during x-ray and fluoroscopic procedures. For example, in intravascular catheter procedures, health care providers may guide the catheter to a target location by using fluoroscopy to track the position of radiopaque markers on the catheter.

Commonly, these radiopaque markers are circumferential metallic bands affixed to the exterior surface of the catheter. Although these marker bands allow the catheter to be visualized by fluoroscopy, they can present certain problems. In particular, metallic marker bands require fixation (e.g., by crimping, swaging, or adhesive) to the underlying catheter to avoid slippage as the catheter is moved through the body. The bands may protrude from the tubular surface of the catheter and increase the catheter profile, which creates frictional resistance to the translational movement of the catheter through body passages, and potentially damages tissues contacting the moving catheter. In some instances, where a marker band has been swaged onto the outer surface of a catheter and the inner diameter of a marker band is greater than the outer diameter of the catheter, buckling may occur, causing the marker band to crack and the catheter surface to tear. Further, the placement of band markers on the outer catheter surface presents problems with inadvertent disassociation of the markers from the catheter wall, with attendant loss of positional and measurement accuracy. In addition, such marker bands are constructed from expensive

2

and heavy radiopaque metals such as gold, platinum, tantalum, and alloys of these dense materials. The use of these heavy materials typically results in inflexible and rigid marker bands that can impair the trackability of the catheter by increasing the stiffness of the catheter, thereby compromising the flexibility and maneuverability of the catheter.

The devices, systems, and methods disclosed herein overcome one or more of the deficiencies of the prior art.

SUMMARY

In one aspect, the present disclosure provides a device for measuring an internal structure in a body of a patient. In one aspect the device comprises an elongate member and a flexible radiopaque coil wound about the elongate member. In some embodiments, the hollow, flexible elongate member has a uniform diameter and includes a proximal portion and a distal portion with a central lumen extending therebetween. In some embodiments, the radiopaque coil includes an alternating series of tightly wound sections and loosely wound sections. In some embodiments, the tightly wound sections comprise areas of greater radiopacity than the loosely wound sections.

In another aspect, the present disclosure provides an imaging system for characterizing and measuring an internal structure in a body of a patient. In some embodiments, the system comprises an elongate member and an imaging device coupled to the elongate member. In some embodiments, the elongate member is hollow and flexible with a uniform diameter. In some embodiments, the imaging system includes a radiopaque coil. In some embodiments, the radiopaque coil is flexible and wound about the elongate member. In some embodiments, the radiopaque coil includes a plurality of tightly wound sections separated by loosely wound sections.

In another aspect, the present disclosure provides a method of characterizing a structure within a body of a patient. In some embodiments, the method comprises inserting an elongate member including a radiopaque marker coil positioned at a distal portion of the elongate member, wherein the radiopaque marker coil has a plurality of evenly spaced radiopaque tightly wound sections separated by linear intervals of constant length. In some embodiments, the method further comprises positioning the radiopaque marker coil of the elongate member adjacent a first point of interest on the structure and observing a first radiopaque tightly wound section at the first point of interest of the structure. The method may further comprise advancing the distal portion of the elongate member through the structure such that first radiopaque tightly wound section is at a second point of interest of the structure; observing a second radiopaque tightly wound section at the first point of interest of the structure; counting the number of intervals separating the first radiopaque tightly wound section and the second radiopaque tightly wound section; and calculating the linear distance between the first point of interest and the second point of interest of the structure by converting the number of intervals separating the first radiopaque tightly wound section and the second radiopaque tightly wound section into a linear measurement.

It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. In that regard, additional

3

aspects, features, and advantages of the present disclosure will be apparent to one skilled in the art from the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings illustrate embodiments of the devices and methods disclosed herein and together with the description, serve to explain the principles of the present disclosure. Throughout this description, like elements, in whatever embodiment described, refer to common elements wherever referred to and referenced by the same reference number. The characteristics, attributes, functions, interrelations ascribed to a particular element in one location apply to those elements when referred to by the same reference number in another location unless specifically stated otherwise.

The figures referenced below are drawn for ease of explanation of the basic teachings of the present disclosure only; the extensions of the figures with respect to number, position, relationship, and dimensions of the parts to form the following embodiments will be explained or will be within the skill of the art after the following description has been read and understood. Further, the exact dimensions and dimensional proportions to conform to specific force, weight, strength, and similar requirements will likewise be within the skill of the art after the following description has been read and understood.

The following is a brief description of each figure used to describe the present invention, and thus, is being presented for illustrative purposes only and should not be limitative of the scope of the present invention.

FIG. 1 is a diagrammatic illustration of an exemplary catheter according to one embodiment of the present disclosure.

FIG. 2 is an illustration of a side view of a distal portion of the exemplary catheter shown in FIG. 1, including an exemplary marker coil according to one embodiment of the present disclosure.

FIG. 3 is a schematic illustration of the exemplary marker coil shown in FIG. 2.

FIG. 4 is an illustration of a perspective view of a portion of an exemplary marker coil according to one embodiment of the present disclosure.

FIGS. 5a-5d illustrate an exemplary method of manufacturing a marker coil on a mandrel according to one embodiment of the present disclosure.

FIG. 6 is an illustration of a cross-sectional view of a distal portion of the exemplary catheter shown in FIG. 1, including a portion of the exemplary marker coil shown in FIGS. 1 and 2.

FIG. 7 is a schematic illustration of an exemplary marker coil within an exemplary catheter according to one embodiment of the present disclosure.

FIG. 8 is an exemplary static fluoroscopy image showing an exemplary marker coil within an exemplary catheter according to one embodiment of the present disclosure.

FIG. 9 is a diagrammatic illustration of an exemplary imaging catheter according to one embodiment of the present disclosure.

FIG. 10 is an illustration of a cross-sectional view of a distal portion of the exemplary catheter shown in FIG. 9, including exemplary inked markers and a portion of an exemplary marker coil.

FIG. 11 is a diagrammatic illustration of a proximal portion of the exemplary imaging catheter shown in FIG. 9, including exemplary inked markers.

4

FIG. 12 is a schematic illustration of exemplary inked markers within an exemplary catheter according to one embodiment of the present disclosure.

FIG. 13 is a diagrammatic illustration of the exemplary imaging catheter shown in FIG. 9, including exemplary inked markers and a portion of an exemplary marker coil.

FIG. 14 is a schematic illustration of a distal portion of the exemplary imaging catheter shown in FIG. 9, including a portion of an exemplary marker coil, an imaging device, and a distal tip.

FIG. 15 is a diagrammatic illustration of a distal portion of the exemplary imaging catheter shown in FIG. 9, including exemplary inked markers.

FIG. 16 is an illustration of the exemplary imaging catheter shown in FIG. 9 positioned within a vessel with the imaging device positioned at a first end of an aneurysm.

FIG. 17 is an illustration of the exemplary imaging catheter shown in FIG. 9 positioned within a vessel with the imaging device advanced to a second end of an aneurysm.

DETAILED DESCRIPTION

For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is intended. Any alterations and further modifications to the described devices, instruments, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one embodiment may be combined with the features, components, and/or steps described with respect to other embodiments of the present disclosure. For the sake of brevity, however, the numerous iterations of these combinations will not be described separately. For simplicity, in some instances the same reference numbers are used throughout the drawings to refer to the same or like parts.

The present disclosure describes devices, systems, and methods to assist health care providers with accurate anatomical structure and/or lesion characterization using external and/or internal imaging. The accurate measurement of anatomical structures and lesions may assist the health care provider in diagnosing a condition, deciding on the appropriate course of treatment, treating the condition, and evaluating the results of the treatment. In one aspect, the present disclosure describes devices, systems, and methods for providing flexible radiopaque markers associated with tubular medical devices such as catheters. In particular, the present disclosure describes a catheter including a radiopaque marker coil having alternating coiled regions of varying pitch and, thus, regions of more and less radiopacity. The marker coil disclosed herein provides an apparatus that can be integrated with tubular medical devices and achieve the radiodensity necessary to be able to visualize and characterize anatomical regions of interest without compromising the flexibility and maneuverability of the device. The marker coil comprises a single length of material that is coiled into tightly wound sections having a closed pitch and loosely wound sections having an open pitch. The tightly wound sections form areas of greater radiopacity while the loosely wound sections form areas of less radiopacity. As the catheter curves to travel through a patient's bodily passages, the marker coil can flex to accommodate for the curvature of

5

the catheter without limiting the flexibility of the catheter. In one aspect, the marker coil is positioned within the catheter wall at a distal portion of the catheter. Moreover, the marker coil disclosed herein may be manufactured more efficiently and at less cost than other radiopaque markers. In another aspect, the present disclosure describes a catheter including the radiopaque marker coil on a distal portion of the catheter as well as inked marker bands on a proximal portion of the catheter. The combination of radiopaque and proportionally spaced inked markers facilitate the ability of the user to accurately estimate internal anatomical structure and lesion measurements.

It should be appreciated that while the exemplary embodiment is described in terms of a catheter, the present disclosure is not so limited. Thus, for example, using the marker coil and/or the inked markers disclosed herein on a tubular medical device such as, by way of non-limiting example, a guidewire or a probe, is within the spirit and scope of the present disclosure.

Referring to FIG. 1, shown therein is a catheter **100** comprising an elongated, flexible tubular member or body **102** including a central lumen **105** that allows the passage of contents from a proximal end **110** through a distal end **115** of the catheter **100**. A radiopaque marker coil **120** is positioned at a distal portion **125** of the body **102**. In general, the catheter **100** is sized and shaped for use within an internal structure of a patient, including but not limited to a patient's arteries, veins, heart chambers, neurovascular structures, gastrointestinal system, pulmonary system, and/or other areas where internal access of patient anatomy is desirable. In that regard, depending on the particular medical application, the catheter **100** is configured for use in cardiology procedures, neurovascular procedures, pulmonary procedures, endoscopy procedures, colonoscopy procedures, natural orifice procedures (such as Natural Orifice Transluminal Endoscopic Surgery (NOTES)), and/or other medical procedures.

The body **102** is shaped and sized for insertion into bodily passages of a human patient. In the pictured embodiment, the body **102** is shaped and configured for insertion into a lumen of a blood vessel (not shown) such that a longitudinal axis **LA** of the catheter **100** aligns with a longitudinal axis of the vessel at any given position within the vessel lumen. In that regard, the straight configuration illustrated in FIG. 1 is for exemplary purposes only and in no way limits the manner in which the catheter **100** may curve in other instances. Generally, the elongate body **102** may be configured to take on any desired arcuate profile when in the curved configuration. In one instance, the body **102** has an overall length from the proximal end **110** to the distal end **115** of at least 90 cm. and in some embodiments, extending to 150 cm. Other lengths are also contemplated. In some instances, the body **102** has an external diameter ranging from 2 F to 9 F (i.e., 0.67 mm to 3 mm).

The body **102** is formed of a flexible material such as, by way of non-limiting example, high density polyethylene, polytetrafluoroethylene, Nylon, block copolymers of polyamide and polyether (e.g., PEBAX), polyolefin, polyether-ester copolymer, polyurethane, polyvinyl chloride, combinations thereof, or any other suitable material for the manufacture of flexible, elongate catheters. In the pictured embodiment, the body **102** is connected at the proximal end **110** to an adapter **130**, which is configured to couple the catheter **100** to another medical device at a proximal port **135** and/or through an electrical connection **137**. Various medical devices that may be coupled to the catheter **100** at the proximal port **135** include, by way of non-limiting

6

example, a storage vessel, a disposal vessel, a vacuum system, a syringe, an infusion pump, and/or an insufflation device. Various devices that may be coupled to the catheter **100** by the electrical connection **137** include, by way of non-limiting example, an energy generator (e.g., an ultrasound generator), a power source, a patient interface module ("PIM"), a computer system, and/or a surgical console.

The lumen **105** is shaped and configured to allow the passage of fluid, cellular material, or another medical device (e.g., a guidewire) from the proximal end **110** to the distal end **115**. In some embodiments, the lumen **105** is sized to accommodate the passage of a guidewire. In such an embodiment, the lumen **105** has an internal diameter greater than 0.014 inches.

The distal end **115** is configured to be inserted into a body cavity, tissue, or tubular organ system of a patient. In some embodiments, the catheter **100** includes a distal tip **140** terminating in the distal end **115**. In some embodiments, the distal tip **140** is tapered to facilitate insertion of the body **102** into a patient. In other embodiments, the distal tip **140** may be blunt, angled, or rounded.

The marker coil **120** is coiled or wound about the distal portion **125** of the body **102**, and has a length **L** extending from a first end **145** to a second end **150**. In various embodiments, the marker coil **120** may have a length **L** ranging from 1 cm to 150 cm. In the pictured embodiment, for example, the marker coil **120** has a length **L** of 24 cm. The marker coil **120** comprises a single length of material that is coiled into tightly wound sections **155** having a closed pitch and loosely wound sections **160** having an open pitch and greater inter-coil spacing than the tightly wound sections. In the pictured embodiment, the marker coil **120** includes 25 tightly wound sections **155**. In other embodiments, the marker coil may have any number of tightly wound sections. The tightly wound sections **155** form areas of greater radiopacity while the loosely wound sections **160** form areas of less radiopacity. Thus, the tightly wound sections **155** effectively form radiopaque markers separated from each other by the loosely wound sections **160**. In one embodiment, the radiopaque material has an external diameter ranging from 0.022 in. to 0.080 inches (i.e., 0.56 mm to 2.03 mm). In some embodiments, the radiopaque material has an external diameter approximating the external diameter of the catheter body.

As shown in FIG. 2, illustrating a section of the distal portion **125** of the body **102**, the marker coil **120** comprises the tightly wound sections **155** separated by the loosely wound sections **160**. As mentioned above, the marker coil **120** is formed of a single length of radiopaque material that has been wound into areas of varying pitch. The radiopaque material may be one or more radiopaque metals including, but not limited to, gold, tungsten, iridium, rhodium, platinum, barium, bismuth, and combinations and/or alloys thereof. However, any material with a high enough radiodensity when shaped into a tightly wound section **155** is suitable for the marker coil **120**. For example, the marker coil **120** may be formed of lower cost alternatives to precious metals with equivalent radiodensity. In some embodiments, the radiopaque material is a radiopaque polymer, which may comprise a matrix of a polymeric material in combination with a radiopaque metal, such as are described above.

As shown in FIGS. 2 and 3, the tightly wound sections **155** are tightly wound areas of the marker coil **120** that form blocks of greater radiopacity or radiodensity than the loosely wound sections **160**. In some embodiments, the tightly wound sections **155** have a width **W** ranging from 1.0 mm

to 2.0 mm. In the pictured embodiment in FIG. 3, the tightly wound sections 155 have a width W of approximately 1.5 mm. Both the tightly wound sections 155 and the loosely wound sections 160 retain the ability to flex, albeit to different degrees. Given the increased flexibility of the radiopaque tightly wound sections 155, the tightly wound sections 155 may have greater widths W (and greater resultant visibility) than rigid, metallic marker bands. Thus, the tightly wound sections 155 form flexible radiopaque markers with spring-like functionality that are capable of curving with the catheter 100 as it traverses through tortuous anatomy without causing the inadvertent catheter kinking and/or trauma that can be caused by rigid marker bands.

The tightly wound sections 155 have a closed pitch while the loosely wound sections have an open pitch. In other words, as shown in FIG. 3, the tightly wound sections 155 are formed of tightly compressed individual coils 162 of the marker coil 120 having little to no space between them, while the loosely wound sections 160 are formed by coils 162 having greater space between centers of adjacent coils 162. In some embodiments, the pitch of the coils 162 in the loosely wound sections 160 may range from 1.1938 mm (0.047 inches) to 1.3462 mm (0.053 inches). In the pictured embodiment, the loosely wound sections 160 are formed by four loosely wound turns or coils 162 of the marker coil 120. However, the loosely wound sections 160 may be formed by any number of coils 162.

In one embodiment, the marker coil 120 is manufactured by stretching a tightly compressed coil at constant intervals past the recovery point of the coil material, thereby creating alternating areas of tightly wound coil and loosely wound coil. Stretching the coil past its recovery point “sets” the intervals between the individual coils (e.g., coils 162) and creates constant intervals between the tightly wound sections 155 and the loosely wound sections 160. For example, in the pictured embodiment in FIG. 3, the tightly wound sections 155 are separated from one another by a constant interval I1, which reflects a fixed distance between adjacent tightly wound sections 155. Similarly, the loosely wound sections 160 are separated from one another by a constant interval I2, which reflects a fixed distance between adjacent loosely wound sections 160. The interval I1 may vary in different embodiments depending upon the particular application desired. For example, in various embodiments, the interval I1 may range from 0.5 cm to 5 cm. In some embodiments, the interval I1 of the marker coil is 1 cm. Stretching the coil past its recovery point to create constant intervals between the tightly wound sections 155 and the loosely wound sections 160, as well as between the individual coils 162, strengthens the marker coil 120.

As shown in FIG. 4, in one embodiment, the marker coil 120 may be manufactured on a mandrel 164. As shown in FIG. 5a, the marker coil 120 may be fixed (e.g., with a fixation device) in an un-stretched condition at the first end 145 about the cylindrical mandrel 164 before being stretched at constant intervals past the recovery point of the coil material, thereby creating alternating areas of tightly wound sections 155 and loosely wound sections 160. In particular, as shown in FIG. 5b, a tightly wound section 155a of the marker coil 120 may be releasably anchored to the mandrel 164 at a coil 165 before the marker coil is stretched in a direction opposite the first end 145 to create a loosely wound section 160a of a desired length L2. The distance between a coil 166 and the coil 165 may be selected based upon a desired width W1 of the tightly coiled section 155a. As shown in FIG. 5c, the marker coil 120 may be releasably anchored to the mandrel 164 at a coil 167 of a tightly wound

section 155b before the marker coil is stretched in a direction opposite the first end 145 to create a loosely wound section 160b of a desired length L3. The distance between a coil 168 and the coil 167 may be selected based upon a desired width W2 of the tightly wound section 155a. In some embodiments, the widths W1, W2 may be substantially the same, but in other embodiments, the widths of individual tightly wound sections may vary in accordance with the desired application for the radiopaque markers. Similarly, in some embodiments, the lengths L2, L3 may be substantially the same, but in other embodiments, the lengths of individual loosely wound sections may vary in accordance with the desired application. As illustrated by FIG. 5d, this process may be repeated until the marker coil 120 comprises the desired number and arrangement of alternating tightly wound sections 155 and loosely wound sections 160. Thus, the marker coil 120 has highly radiopaque tightly wound sections 155 alternating with less radiopaque loosely wound sections 160 at substantially constant intervals, allowing the catheter 100 to serve as an internal marking or measuring device.

In the embodiment shown in FIG. 6, the marker coil 120 is at least partially enclosed within an outer wall 170 of the catheter body 102. The outer wall 170 extends from an inner surface 172 to an outer surface 174. In the pictured embodiment, the inner surface 172 forms a luminal surface of the catheter 100 adjacent the lumen 105. In at least one embodiment, the marker coil 120 is completely enclosed between the inner surface 172 and the outer surface 174. In some embodiments, an edge or end of the marker coil may be exposed through the outer surface 174 and/or the inner surface 172 of the outer wall 170. For example, in some embodiments it may be desirable for an edge or end of the marker coil 120 to protrude from the wall 170 to anchor itself or other catheter components (e.g., a balloon or a stent) to the body 102.

The marker coil 120 can be placed within and/or about the outer wall 170 in accordance with a variety of methods. In the pictured embodiment, the outer wall 170 includes a lumen 176 sized and configured to receive the marker coil 120 during manufacture of the catheter 100 without increasing the outer diameter D or profile of the body 102 of the catheter 100. The lumen 176 comprises an annular space extending between the inner surface 172 and the outer surface 174. The space within the lumen 176 allows the radiopaque marker coil 120 to bend and flex freely beneath the outer wall 170. In some embodiments, the marker coil 120 may be wound directly onto the catheter (i.e., into the inner surface 172) under tension before the incorporation of the outer surface 174 to form a series of flexible radiopaque markers that are embedded within the outer wall 170. For example, the outer surface 174 of the outer wall 170 may be applied as a polymeric solution over the marker coil 120, or as a heat-shrink film that is wrapped around the coil 120.

In some instances, such as where the radiopaque material of the marker coil 120 is a metal wire, the material of the outer wall 170 is softer than the radiopaque material of the marker coil. In such instances, the marker coil 120 may be directly embedded into the outer wall 170 by sliding the marker coil over the distal portion 125 of the body 102 and directly apply force to the marker coil to push it into the material of the outer wall. For example, in some embodiments, a compressive apparatus (e.g., a roller assembly) may be used to physically press the marker coil 120 circumferentially and along its length into the outer wall 170 of the catheter 100. In some instances, the outer surface 174 may include indentations, grooves, or other surface features

shaped and configured to receive the marker coil **120** without disrupting the inner surface **172** or increasing the outer diameter **D** or profile of the body **102** of the catheter **100**. In some embodiments, the marker coil **120** is incorporated into the outer wall **170** of the catheter **100** in such a manner as to maintain a smooth outer surface of the catheter **100**. Such a smooth surface has improved lubricity over a comparable catheter having metallic marker bands disposed around the outer wall **170**.

FIG. **7** is a schematic illustration of the marker coil **120** enclosed within an outer wall **170** of the body **102** of the catheter **100**. In some embodiments, the marker coil **120** acts as a support mechanism to the wall **170** of the catheter **100** and lends an additional degree of stiffness to the distal portion **125** of the body **102** of the catheter **100**. As a result, the distal portion **125** may be provided with greater longitudinal stiffness, which may improve the device pushability of the catheter **100**.

The method of stretching a single coil to form the more radiopaque, tightly wound sections **155** separated by constant intervals **I1** creates a bound series of radiopaque markers, and the method of integrating the unitary marker coil **120** with the catheter **100** avoids the need to individually place radiopaque markers along a medical device. The embodiments disclosed herein eliminate the need to individually place or swage separate radiopaque markers onto a medical device. Instead, the unitary marker coil **120** comprises several constantly spaced tightly wound sections **155** or radiopaque markers that may be integrated into the body **102** of the catheter **100** without disturbing their predetermined distances from one another. Therefore, the embodiments disclosed herein avoid the positioning errors associated with the discrete placement of individual radiopaque markers (e.g., marker bands). Moreover, the embodiments disclosed herein allow for a less expensive and possibly less time-consuming manufacturing process than that required by the discrete placement of individual radiopaque markers.

FIG. **8** illustrates a fluoroscopic or X-ray image **400** showing the catheter **100** positioned within a patient. In particular, the distal portion **125** of the catheter **100** is shown extending through the coronary vasculature. The tightly wound sections **155** of the marker coil **120** are prominently visible on the image **400**, while the less radiopaque loosely wound sections **160** are less prominently highlighted on the image **400**. A healthcare provider may utilize the marker coil **120** to evaluate, localize, and measure anatomical structures and lesions as described below in relation to FIGS. **16** and **17**. In addition, the healthcare provider may draw conclusions about the three-dimensional structure or plane of curvature of the intraluminal structure and/or lesion by observing the relative distances between adjacent tightly wound sections **155**. For example, shortened or absent intervals **I1** between adjacent tightly wound sections **155** on the image **400** may indicate an out-of-plane deflection of the catheter **100**. In other words, shortened or absent intervals **I1** between adjacent tightly wound sections **155** on the image **400** may indicate curvature of the catheter **100** in a third dimension or a z-plane. Such data could be used in combination with angiography and/or intravascular imaging to generate a three-dimensional representation of the marker coil **120** and/or the intraluminal structure and/or lesion where the marker coil is located.

Referring back to FIG. **1**, in some embodiments, the catheter **100** includes an imaging device, such as, by way of non-limiting example, an intravascular ultrasound (“IVUS”) transducer **200**. Thus, in some embodiments, the catheter **100** may comprise an IVUS catheter. In such embodiments,

the catheter **100** may be connected at the electrical connection **137** to an IVUS imaging system. In the pictured embodiment, the distal tip **140** houses the transducer **200**. The marker coil **120** provides radiopaque markers in the form of the tightly wound sections **155** to assist in positioning the transducer **200** within a patient’s vasculature and obtaining accurate visualization and measurements of the patient’s vessels. In some instances, the imaging device may be used to determine the morphology and pathology of a target lesion within a patient’s anatomy (e.g., a restriction within a vessel). The radiopaque tightly wound sections **155** allow for the accurate localization and measurement of such a lesion. Intraluminal imaging may be done as an initial step to help determine the best applicable therapy, to observe a therapeutic measure in real-time, or as a later step to assess the results of a given therapy.

It should be appreciated that while the exemplary embodiment is described in terms of an ultrasonic device, to render images of a vascular object, the present disclosure is not so limited. It should be noted that the catheter **100** depicted herein is not limited to a particular type of device, and includes any of a variety of imaging devices. Thus, for example, using backscattered data (or a transformation thereof) based on other sources of energy, such as electromagnetic radiation (e.g., light waves in non-visible ranges such as used in Optical Coherence Tomography, X-Ray CT, spectroscopy, etc.), to render images of any tissue type or composition (not limited to vasculature, but including other structures within a human or non-human patient) is within the spirit and scope of the present disclosure.

With reference to FIG. **1**, in some embodiments, the catheter **100** includes an expandable device **300**, such as, by way of non-limiting example, a balloon apparatus and/or a stent. The expandable device **300** may be positioned about the distal tip **140** and/or the distal portion **125**. The expandable device **300** may be positioned around the coil marker **120** such that the radiopaque portions of the coil marker **120** can be visualized on fluoroscopy through the expandable device as the catheter **100** is maneuvered through the patient’s body. The marker coil **120** provides radiopaque markers in the form of the tightly wound sections **155** to assist in positioning the expandable device **300** within a patient’s vasculature and obtaining accurate visualization and measurements of the patient’s vessels. In some embodiments, the length of the expandable device **300** is at least as long as the length **L** of the coil marker **120** extending from the first end **145** to the second end **150**.

In some instances, the catheter **100** is used to deliver a medical device such as a balloon, a stent, a graft, a stent-graft, a vena-cava filter, or other implantable medical device, hereinafter collectively referred to as the expandable device **300**. For example, the expandable device **300** may comprise a self-expanding stent or may comprise a balloon used to deliver and/or deploy a balloon-expandable stent. In some instances, the expandable device **300** comprises a drug-eluting device such as a drug-eluting balloon or a drug-eluting stent. The radiopaque markers provided by the tightly wound sections **155** may assist in the precise intravascular delivery and deployment of the expandable device **300**. In particular, the expansion of the expandable device **300** may be monitored using fluoroscopy-mediated visualization of the radiopaque tightly wound sections **155**. In some instances, the expandable device **300** is positioned about the wall **170** of the catheter **100** in such a manner as to ensure that the expandable device **300** does not extend substantially beyond the length **L** of the marker coil **120**.

11

Due to the non-linear nature of many body lumens, it is often desirable to measure the length of a body lumen along a center line extending axially through the center of the body lumen. Such measurements can be useful in selecting an appropriately sized luminal implant or prosthesis. In that regard, in some instances, the expandable device **300** comprises a positioning element shaped and configured to center the catheter **100** within a lumen (e.g., a curved lumen) so as to facilitate the accurate measurement of an intraluminal structure or lesion. Such an expandable device may have a diameter in an expanded state that at least corresponds to the diameter of the body lumen. In this way, the expandable device **300** can be expanded to center the catheter **100** within the body lumen, thereby allowing the length of the lesion (or area of interest within the lumen) to be measured along the center of the body lumen.

FIG. **9** shows an imaging catheter **500** including the marker coil **120** and a plurality of inked markers **505** according to one embodiment of the present disclosure. The imaging catheter **500** is substantially similar to the catheter **100** except for the differences noted herein. The catheter **500** comprises an elongate, tubular member or body **510** extending from a proximal end **515** to a distal tip **525** that terminates at a distal end **520**, an imaging device **530** disposed on an imaging housing **532**, and an adapter **130** coupling the body **510** to a PIM **535**.

The body **510** is substantially similar to the body **102** of the catheter **100** except for the differences noted herein. The body **510** includes a proximal shaft portion **540**, which includes the plurality of inked markers **505**, and a distal shaft portion **545**, which includes the marker coil **120**. As shown in FIG. **10**, the body **510** comprises an inner tubular member **550** disposed within an outer tubular member **555**. The inner tubular member **550** extends the length of the body **510** and is coupled at either end to the adaptor **130** and the distal tip **525** (shown in FIG. **9**). In some embodiments, the inner tubular member **550** is heat bonded to the adapter **130** and the distal tip **525**. However, any of a variety of coupling methods may be employed to secure the inner tubular member **550** to the adaptor **130** and the distal tip **525**.

In the pictured embodiment, the inner tubular member **550** comprises a darkly colored, elongate, cylindrical tube. The inner tubular member **550** defines a lumen **560** extending the length of the body **510** from the adaptor **130** to the distal tip **525**. The lumen **560** is substantially identical to the lumen **105** described above in relation to FIG. **1**.

The outer tubular member **555** comprises a clear cylindrical sleeve that extends from the adapter **130** to the imaging device housing **532**. In some embodiments, the outer tubular member **555** is secured to the adapter **130** and the imaging device housing **532** by means of an adhesive. However, any of a variety of coupling methods may be employed to secure the outer tubular member **555** to the adaptor **130** and the imaging device housing **532**. The outer tubular member **555** includes a smooth outer surface **556** configured to reduce the amount of friction created on an introducer device during insertion and removal of the catheter **500**, thereby facilitating smooth and accurate pullbacks during imaging procedures.

As shown in FIG. **10**, the plurality of inked markers **505** are disposed on the inner tubular member **550**. The inked markers **505** comprise direct visualization markers that may be viewed during use with the naked eye (or with the use of an endoscope), in contrast to the tightly wound sections **155** of the marker coil **120**, which comprise radiopaque markers. The inked markers **505** comprise lightly-colored markings on an exterior surface **562** of the inner tubular member **550**.

12

The lightly-colored inked markers **505** appear clearly against a background of the darkly-colored inner tubular member **550** and can be visualized easily through the clear outer tubular member **555** with and without the use of light. In alternate embodiments, the inner tubular member may be light-colored while the inked markers are darkly-colored, provided there is adequate contrast between the colors of the inner tubular member and the inked markers to permit visualization through the outer tubular member **555**. For example, in some embodiments, the inner tubular member **550** may have a dark color, such as black, dark blue, dark grey, or the like, while the inked markers **505** have a light color such as white, light blue, light green, pink, or the like. The high contrast between the plurality of inked markers **505** and the inner tubular member **550** facilitates visualization of the markers in a low light environment, such as a darkened operating room.

The inked markers **505** may be formed of a variety of suitable inks, which are typically indelible. Because the inked markers **505** are shielded from the patient's anatomy by the outer tubular member **555**, the ink does not necessarily have to be biocompatible. In some embodiments, the inked markers **505** may be formed of a fluorescent substance, thereby enabling the inked markers to be more clearly visualized in a low light environment, such as a darkened operating room. The inked markers **505** can be applied to the inner tubular member **550** by any of a variety of suitable methods, including, without limitation, painting, spraying, masked dipping (i.e., dipping the body **510** in ink with parts of the body masked to avoid being coated by the ink), and electrostatic attraction.

FIG. **11** illustrates a portion of the proximal shaft portion **540**, which includes the plurality of inked markers **505**. In the pictured embodiment, the plurality of inked markers **505** comprise markers of different widths. In particular, the plurality of inked markers **505** comprise inked markers **563**, which have a width **W3**, and inked markers **564**, which have a width **W4**. In various embodiments, the widths of the individual inked markers may be uniform or different, depending upon the desired application of the inked markers **505** and/or the catheter **500**. The widths of the individual inked markers may range from 0.5 mm to 5.5 mm. In at least one embodiment, the width **W3** is approximately 1.0 mm and the width **W4** is approximately 5.0 mm.

FIG. **12** illustrates the proximal shaft portion **540** of the catheter **500**, including the inked markers **563**, **564**. In other embodiments, any number of inked markers **563** and the inked markers **564** may be positioned on the inner tubular member **550** in any of a variety of combinations or arrangements. In some embodiments, the inked markers **563** and **564** may have the same width, comprising a plurality of equally-sized inked markers. In some embodiments, the inked markers **563** may have a different color than the inked markers **564** to facilitate distinguishing the markers **563**, **564** from one another during use. In the pictured embodiment in FIG. **13**, the inked markers **563** and the inked markers **564** are arranged in a repeating pattern of four inked markers **563** and one inked marker **564**. The inked markers **563** and the inked markers **564** are separated from one another by an interval **I3**. The interval **I3** may vary in different embodiments depending upon the particular application desired. The interval **I3** may range from 3.0 mm to 20.0 mm. For example, in the pictured embodiment in FIG. **11**, the interval **I3** measures approximately 1 cm, which reflects a fixed distance between adjacent inked markers **505**.

Returning to FIG. **10**, the outer tubular member **555** circumferentially and longitudinally surrounds the inner

13

tubular member 550, creating an annular space or lumen 565 therebetween. In the pictured embodiment, the marker coil 120 is at least partially enclosed in the lumen 565 between the inner tubular member 550 and the outer tubular member 555 at the distal shaft portion 545. In at least one embodiment, the marker coil 120, including the tightly wound sections 155 and the loosely wound sections 160, is completely enclosed in the lumen 565 at the distal shaft portion 565. In the pictured embodiment of FIG. 9, the marker coil 120 comprises twenty-five tightly wound sections 155 separated from each other by approximately 100 mm.

FIGS. 14 and 15 show the distal shaft portion 545 of the catheter 500, including the imaging device 530 positioned proximal to the distal tip 525 at the imaging device housing 532. The distal tip 525 is configured to be inserted into a body cavity, tissue, or tubular organ system of a patient. In the illustrated embodiment, the distal tip 525 is tapered to facilitate insertion of the body 510 into a patient. As shown in FIG. 15, the distal tip 525 has a length LA ranging from 10 mm to 20 mm. In various embodiments, the length L4 varies depending upon the particular application of the catheter 500. For example, in the pictured embodiment, the length L4 is approximately 12 mm.

The image device housing 532 has a length L5 ranging from 1.5 mm to 10.0 mm. In various embodiments, a length L5 varies depending upon the particular type of imaging device 530 and the particular application of the catheter 500. In the pictured embodiment, the imaging device 530 comprises an ultrasound transducer (e.g., by way of non-limiting example, a solid state phased array, a rotational, FLIVUS, and/or PMUT transducer). However, in other embodiments, the imaging device 530 may comprise any of a variety of imaging devices, including, by way of non-limiting example, an OCT or laser-emitting device. For example, in the pictured embodiment where the imaging device 530 comprises an ultrasound transducer, the length L5 is approximately 6.5 mm.

In the pictured embodiment in FIG. 15, the tightly wound sections 155 of the marker coil 120 are separated from each other and the imaging device 530 by an interval I1, as described above in relation to FIG. 3. The interval I1 may vary in different embodiments depending upon the particular application desired. For example, in the pictured embodiment, the interval I1 of the marker coil 120 is 1 cm.

FIGS. 16 and 17 illustrate an exemplary instance where the catheter 500, with its combination of proximally-positioned inked markers 505 and its distally-positioned marker coil 120, can be used to assist the healthcare professional with accurate and efficient lesion measurement and evaluation. In FIG. 16, the distal shaft portion 545 of the catheter 500 is positioned within a vessel 600, which includes an intravascular lesion 605. In the pictured embodiment, the vessel 600 comprises an aorta and the intravascular lesion comprises an aortic aneurysm, which is an abnormal dilation of the aorta. However, the catheter 500 (or the catheter 100 described above) may be used to evaluate a variety of lesions and/or structures, as described further below.

In FIG. 16, the distal shaft portion 545 of the catheter 500 is shown positioned within the vessel 600 such that the marker coil 120 is located adjacent to the lesion 605. In particular, the distal-most tightly wound section 155c is positioned immediately proximal the lesion 605, which may be visualized with contrast during fluoroscopy, without contrast during fluoroscopy (i.e., by visualizing the radiopaque tightly wound sections 155), and/or with the use of the imaging device 530. During fluoroscopy, the tightly wound sections 155 of the marker coil 120 remain visible,

14

thereby allowing the healthcare provider to position the marker coil 120 relative to margins of the lesion 605 to measure various dimensions of the lesion 605. In some instances, the healthcare professional may visualize the lesion 605 using fluoroscopy to accurately position the radiopaque tightly wound section 155c at a first end 606 of the lesion 605. In some instances, the healthcare professional may use the imaging device 530 to localize the lesion 605 and accurately position the radiopaque tightly wound section 155c at the first end 606 of the lesion 605. For example, the imaging device 530 may be used to localize relevant anatomical landmarks and/or the margins of the lesion 605. In some instances, the healthcare professional may use a combination of extravascular imaging (i.e., fluoroscopy) and intravascular imaging (i.e., with the imaging device 530) to localize the lesion 605 and accurately position the radiopaque tightly wound section 155c at the first end 606 of the lesion 605.

Intravascular imaging can occur before or after interventional treatment. When used before such treatments, the images may aid the healthcare profession in localizing the lesion 605 and in deciding whether and/or how to treat the lesion. For example, in the case of aneurysms, the imaging device 540 may be used to evaluate the health of the tissue wall prior to selecting an implanting a stent graft to repair the aneurysm and selecting anchor points for the stent/graft. Moreover, the imaging device 540 may be used to determine both the position and orientation of the tightly wound sections 515 (i.e., as the distal shaft portion 545 is curved to approximate the boundaries of the lesion 605) to assist in evaluating the lesion 605. For example, the imaging device 540 may be utilized to determine position of individual tightly wound sections 515 relative to each other to evaluate the shape and/or curvature of the lesion 605. The resulting data may be utilized to construct a three dimensionally significant representation of the lesion 605 as indicated by the positions of the tightly wound sections 515. Such a representation could yield a length, diameter, and/or radius of curvature of a lesion 605, all of which could be used to assist in the selection of an appropriately sized prosthetic or implant, such as a stent graft for repairing an aneurysm.

When used after a given treatment, the images may aid the healthcare professional in the assessment and documentation of the results of the treatment. Utilizing intravascular imaging from the imaging device 540 in combination with the radiopaque tightly wound sections 155 of the marker coil 120 enables the healthcare provider to use less contrast during the localization and the measurement of the lesion 605. Moreover, the positions of the radiopaque tightly wound sections 155 observed with fluoroscopy may be co-registered with the positions of the tightly wound sections 155 observed with intravascular imaging to enhance the evaluation, localization, and measurement of the lesion 605.

Once the distal-most tightly wound section 155c is positioned at the first end of the lesion 605, the healthcare provider can note the relevant inked markers 505 as described below and perform whatever diagnostic and/or therapeutic measures may be indicated for the particular procedure he or she is performing. Thereafter, the healthcare provider can advance the catheter 500 into the vessel 600 until the tightly wound section 155c is positioned at a second end 607 of the lesion 605.

In FIG. 17, the distal-most tightly wound section 155c is positioned at a second end 607 of the lesion 605, which may be visualized with contrast during fluoroscopy, without contrast during fluoroscopy (i.e., by visualizing the

15

radiopaque tightly wound sections 155), and/or with the use of the imaging device 530 as described above. The healthcare provider can use the marker coil 120 to estimate the length and/or other dimensions of the lesion 605. In some instances, the healthcare provider can observe which particular tightly wound section 155d is positioned at the first end 606 of the lesion 605 when the distal-most tightly wound section 155c is positioned at the second end 607 of the lesion 605. By comparing the tightly wound section 155c to the tightly wound section 155d, the healthcare provider may estimate the length of the lesion 605. In particular, by counting the number of intervals I1 between the tightly wound section 155c and the tightly wound section 155d, and correlating that number of intervals to a length measurement, the healthcare provider can estimate the length of the lesion 605. As described above, the spacing between the tightly wound sections 155 can be of a constant length interval I1 so that the healthcare professional can convert the difference in the number of intervals into a corresponding length measurement. For example, in the example shown in FIGS. 16 and 17, there are nine intervals I1 between the tightly wound section 155c and the tightly wound section 155d. Given that each interval I1 of the marker coil 120 measures 100 mm or 1 cm, the length of the lesion 605 is approximately 900 mm or 9 cm. Thus, the multiple tightly wound sections 155 of the marker coil 120 provide several radiopaque markers that allow for more accurate measurement of various dimensions of an anatomical structure and/or lesion than a measuring device having only a few radiopaque markers.

The plurality of inked markers 505 allow the healthcare professional to visually detect how far the catheter 500 is axially moved with respect to a reference point such as a proximal end 610 of an introducer 615. At least some of the proximal shaft portion 540 of the catheter 500 is positioned outside the patient's body, allowing the healthcare professional to estimate the length of the catheter 500 that has been inserted into the patient's body by observing the number of inked markers 505 still visible outside the patient's body. In addition, the healthcare professional may estimate the length of the intravascular lesion 605 by noting a first number of externally visible inked markers 505 when the catheter 500 is positioned at the first end 606 of the lesion 605 as shown in FIG. 16, noting a second number of externally visible inked markers 505 when the catheter 500 is positioned at the second end 607 of the lesion 605 as shown in FIG. 17, and comparing the two values. In particular, the difference between the first number and the second number can be used to estimate the length of the lesion 605. As described above, the spacing between the inked markers 505 can be of a constant length interval so that the healthcare professional can convert the difference in the number of inked markers into a length measurement corresponding to how far the catheter 500 has axially moved with respect to a reference point such as a proximal end 610 of an introducer 615.

In another instance, the healthcare professional can observe a particular inked marker 505a present at a reference point outside the patient's body, such as the proximal end 610 of the introducer 615, when the catheter 500 is positioned at the first end 606 of the lesion 605 as shown in FIG. 16, observe a particular inked marker 505b present at the same reference point when the catheter 500 is positioned at the second end 607 of the lesion 605 as shown in FIG. 17, and comparing the length measurements indicated by the inked markers 505a, 505b. Each inked marker 505 is positioned a particular distance from the proximal end 515 of the catheter 500. For example, if the inked marker 505a is

16

positioned 30 cm from the proximal end 515 and the inked marker 505b is positioned 22 cm from the proximal end 515, the healthcare professional may estimate that the length of the lesion 605 is 8 cm.

Once the lesion 605 has been measured, the healthcare provider can perform whatever diagnostic and/or therapeutic measures may be indicated for the particular procedure he or she is performing. For example, in some embodiments, the healthcare professional may advance the catheter 500 into the lesion 605 and image the lesion 605 using the imaging device 530. In other instances, the healthcare professional may expand an expandable device similar to the expandable device 300 shown in FIG. 1. Knowing the accurate length and/or other dimensions of the lesion 605 will help the healthcare provider to determine the appropriate course of treatment, appropriately evaluate the results of a given treatment, and/or appropriately position any implantable devices for treating the lesion 605. For example, in the situation illustrated by FIGS. 16 and 17, the healthcare provider may select an appropriately sized stent and expand the stent within the lesion 605 (i.e., an aneurysm) within the vessel 600.

Embodiments in accordance with the present disclosure provide users with an accurate and efficient device, system, and method for evaluating, localizing, and measuring anatomical structures and/or lesions to plan an appropriate treatment course and/or evaluate a given treatment. Proximal inked markers and/or a distal radiopaque marker coil in accordance with the present disclosure may be used alone or in combination in a variety of applications to evaluate, localize, and measure anatomical structures and/or lesions. For example, but not by way of limitation, embodiments of the present disclosure may be used to assist in evaluating an intraluminal site for implantation of a prosthesis (e.g., a drug-eluting balloon, a drug-eluting stent, a stent graft, a bioresorbable stent), PTCA balloon location, an endovascular aneurysm repair (e.g., of the abdominal or thoracic aorta), IVC filter placement (e.g., in the inferior vena cava), evaluation of tumor growth/response to treatment, and a variety of other procedures previously utilizing a separate measuring device (e.g., a ruler) for lesion measurement. In addition, embodiments of the present disclosure may be used to monitor the position and/or efficacy of already implanted devices such as, but not by way of limitation, stents, stent grafts, drug-eluting stents, drug-eluting balloons, and orthopedic implants (e.g., bone screws or hip, shoulder, or knee implants). Moreover, the markers of the present disclosure enable the elimination of at least one pigtail marker catheter exchange in several procedures, such as a low contrast endovascular aneurysm repair procedure, thereby accelerating the process of stent-graft length sizing assessment. Some embodiments of the present disclosure may be used in a variety of organ systems such as, but not by way of limitation, the circulatory system, the lymphatic system, the digestive system, the pulmonary system, the orthopedic system, and the neurological system.

Persons of ordinary skill in the art will appreciate that the embodiments encompassed by the present disclosure are not limited to the particular exemplary embodiments described above. In that regard, although illustrative embodiments have been shown and described, a wide range of modification, change, and substitution is contemplated in the foregoing disclosure. It is understood that such variations may be made to the foregoing without departing from the scope of the present disclosure. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the present disclosure.

17

What is claimed is:

1. An apparatus comprising:
 - a catheter defining a lumen; and
 - a radiopaque wire comprising a single piece of material arranged around the lumen of the catheter to form:
 - a first section and a third section that each comprises:
 - a same first length; and
 - a tightly wound portion of the single piece of material forming a radiopaque marker; and
 - a second section and a fourth section that each comprises a same second length that is longer than the first length,
 - wherein the second section is positioned between the first section and the third section such that the first section and the third section are spaced from one another by the second length,
 - wherein the third section is positioned between the second section and the fourth section, such that the second section and the fourth section are spaced from one another by the first length.
2. The apparatus of claim 1, wherein the radiopaque wire comprises:
 - a fifth section that comprises the first length and the tightly wound portion of the single piece of material forming the radiopaque marker; and
 - a sixth section that comprises the second length.
3. The apparatus of claim 2,
 - wherein the fourth section is positioned between the third section and the fifth section such that the first section, the third section, and the fifth section are evenly spaced from one another by the second length, and
 - wherein the fifth section is positioned between the fourth section and the sixth section such that the second

18

- section, the fourth section, and the sixth section are evenly spaced from one another by the first length.
- 4. The apparatus of claim 1, wherein the second section and the fourth section each comprises a loosely wound portion of the single piece of material that does not form the radiopaque marker.
- 5. The apparatus of claim 1, wherein the radiopaque wire is embedded within a wall of the catheter.
- 6. The apparatus of claim 1,
 - wherein the catheter comprises a tubular member, and
 - wherein the radiopaque wire is disposed around an outer surface of the tubular member.
- 7. The apparatus of claim 1, wherein the radiopaque wire is disposed at a distal portion of the catheter.
- 8. The apparatus of claim 1, further comprising an imaging device disposed at a distal portion of the catheter.
- 9. The apparatus of claim 8, wherein the imaging device is configured for intravascular ultrasound (IVUS) or optical coherence tomography (OCT).
- 10. The apparatus of claim 1, further comprising a therapeutic device disposed at a distal portion of the catheter.
- 11. The apparatus of claim 10, wherein the therapeutic device comprises at least one of a balloon or a stent.
- 12. The apparatus of claim 1,
 - wherein the catheter comprises:
 - an inner tubular member defining the lumen; and
 - an outer tubular member surrounding the inner tubular member, and
 - wherein the radiopaque wire is positioned inside a space between the inner tubular member and the outer tubular member.

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